

Polymarket Trader Analysis

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September 28, 2025

1 Introduction

Given this dataset of 600,000 Polymarket traders, I wanted to figure out how to make money from it. However, while examining the data, I noticed that this wasn't time-series data where I could just predict what happens next. Instead, it was a snapshot of trader performance. Therefore, I concluded that I had to figure out what makes successful traders so successful and then copy their approach.

2 Part 1: Started with Correlations

The first thing I did was run correlations on everything to see what actually matters for making money.

Feature	Correlation (r)
trader_volume	0.4076
transaction_count	0.1980
transactions_per_day	0.0605
markets_per_day	0.0113
mean_tx_value	0.0101

Table 1: Correlations with trader_pnl

Volume had the highest correlation at 0.41, which seemed like the obvious answer. But I knew it couldn't just be this obvious, so I looked more into it by splitting everyone into volume deciles to see who actually makes money.

Volume Decile	Average PnL (\$)
0 (Lowest)	-0.09
1	-0.88
2	-2.06
3	-1.51
4	-2.69
5	-4.82
6	-5.69
7	-11.17
8	-35.65
9 (Top 10%)	+64.54

Table 2: Average PnL by Volume Decile

So volume is basically useless unless you already had a lot of it. The other 90% of traders need a different approach. This meant I had to find skill-based edges instead of capital-based ones.

3 Part 2: Hypothesis Tests

I then looked at the next thing that seemed obvious to me to affect PnL: Specialization. I tested this by looking at how specialized people are (by measuring by their largest topic share) and whether that correlates with profits.

Specialization Decile	Average PnL (\$)
0 (Most Diversified)	+1.39
1	+165.52
2	+63.58
3	-20.36
4	-14.14
5	-65.12
6 (Most Specialized)	-32.72

Table 3: Average PnL by Specialization Level

Through this data, I determined that moderate diversification is more profitable than hyper-specialization. The people who focus on just one thing actually lose money.

I also looked at which topics make money to see if some areas are easier to make money in than others.

Primary Topic	Mean PnL (\$)
Science and Technology	+27.43
Sport	+22.52
Religion	+18.08
Conflict, War and Peace	+1.55
Politics	-18.98
Economy, Business and Finance	-45.41
Education	-152.62

Table 4: Profitability by Primary Topic

Here, I concluded that the optimal strategy included not only should diversifying, but picking the right topics. Sports and science/tech make money, while economics and education lose money.

4 Part 3: Market Rebalancing

At this point, I knew successful traders were moderately diversified, but I didn't know what they were actually doing to make money. So I went back to the correlation data and noticed that high-PnL traders had some specific characteristics:

- They trade a lot every day (high transactions per day)
- Their trades don't last very long (low `std_time`)
- They cash out quickly and consistently
- They consume a lot of price levels but don't impact the market much per trade

Based on my earlier research into how people gain money in markets, this behavior looked like market rebalancing to me.

So I created a formula to measure this against PPV since PPV tells you how much a trader makes for every dollar they put through the market, which makes it a cleaner measure of execution quality. I also added `std_time` in the denominator, because faster exits and shorter holding periods are common features of rebalancing.

$$\text{market_rebalancing_score} = \frac{\text{transactions_per_day}}{1 + \text{price_levels_vw_per_transaction}} \times \frac{\text{trader_ppv}}{1 + \text{std_time}} \quad (1)$$

When I tested this against PPV in a correlation test, I got a score of 0.2708.

5 Part 4: Archetype's Using ML

I now had two findings:

1. Moderate diversification beats specialization
2. Market rebalancing is profitable

But I wanted to see if these combine. To test this, I used machine learning and used six features that I thought capture trader behavior:

$$\text{Features} = \{\log_volume, \text{largest_transformers_topic_share}, \quad (2)$$

$$\text{transactions_per_day, markets_per_day}, \quad (3)$$

$$\text{std_tx_value, price_levels_per_transaction}\} \quad (4)$$

Then I ran K-means clustering with $k = 4$ to find distinct trader types. Here's what came out:

Archetype	Count	Mean PnL	Med Vol	Med Tx	Specialization
Active Generalists	19,699	+467.51	257.00	23.00	0.51
Occasional Explorers	254,329	-4.16	145.99	10.00	0.53
Patient Specialists	244,211	-4.98	58.00	2.00	1.00
High-Conviction Gamblers	86,339	-80.33	5,817.72	26.00	0.73

Table 5: Trader Archetypes and Performance

Archetype Definitions:

Active Generalists: Trade a lot across multiple topics and get out fast; they stack small, consistent edges instead of swinging for home runs.

Occasional Explorers: Trade rarely with small tickets and longer holds; results look more like noise than a repeatable process.

Patient Specialists: Stick to one lane and wait; when they're right it lands in clumps, but carry and execution costs eat a lot in between.

High-Conviction Gamblers: Make big, concentrated bets and move the market; they pay for immediacy and live with very bimodal outcomes.

The Active Generalists crush everyone else with a mean PnL of +\$467.51, which supports my hypothesis. I ran t-tests to make sure the results were statistically significant:

- Active Generalists vs Occasional Explorers: $p\text{-value} = 1.097\text{e-}03$
- Active Generalists vs Patient Specialists: $p\text{-value} = 1.076\text{e-}03$
- Active Generalists vs High-Conviction Gamblers: $p\text{-value} = 3.210\text{e-}04$

6 Part 5: Final Strategy

1. Be an Active Generalist

- Trade moderately frequently
- Diversify across multiple topics but focus 60–70% on profitable ones
- Don't go crazy with position sizes

2. Do Market Rebalancing

- Use patient limit orders to capture bid-ask spreads
- Target quick profits rather than big directional bets
- Keep `mean_delta` above 0.3 to ensure you're actually capturing edge

The best part about this is it works without needing a lot of capital. The key insight is that sustainable profits come from combining moderate diversification with good execution, not from having the most money or knowing the most about one topic.

One additional study I wanted to run (but didn't know how to) was testing which topics are actually profitable and how aggressively they were traded to generate those profits. Since the dataset includes the `topic_*` columns and each trader's `largest_transformers_topic_share`, I could estimate the proportion of volume allocated to a specific topic. Combining this with each trader's PPV would let me estimate how profitable their focus areas really were. I also wanted to use `price_levels_consumed` as a way to estimate how aggressively they priced their trades.